COMPUTER ENGINEERING

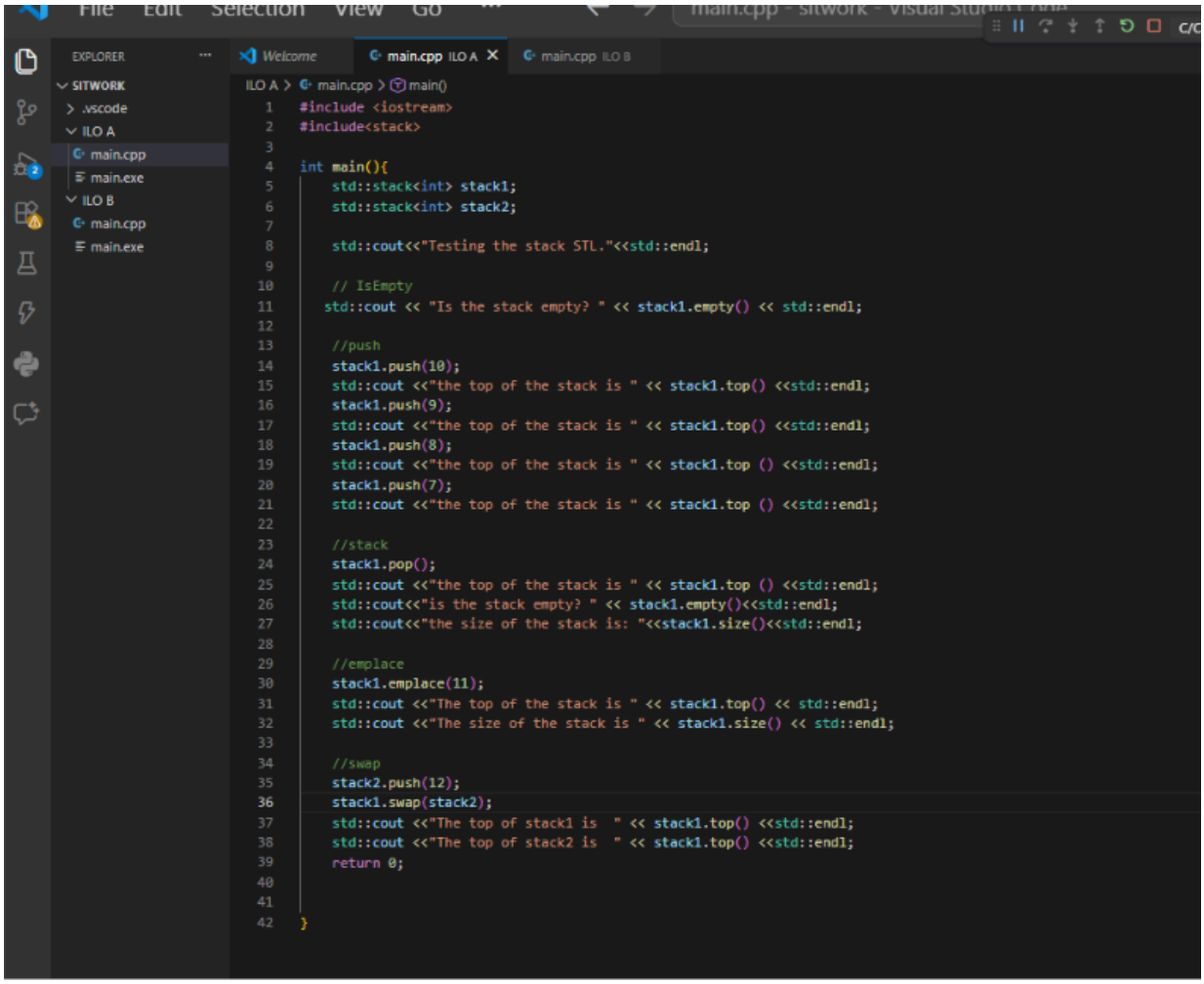
QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structure and Algorithm	**SECTION:**	CPE21S2
DATE PERFORMED:	07/30/26	**DATE SUBMITTED:**	07/30/26
NAME:	Nieles, Cenndy M.	**INSTRUCTOR:**	Engr.Jim Lord Quejado

ACTIVITY NO. 4.1___ **Seatwork 4.1: Array-Based vs. STL Stack Implementations**

1. PROCEDURE OUTPUT

What procedures did you perform?



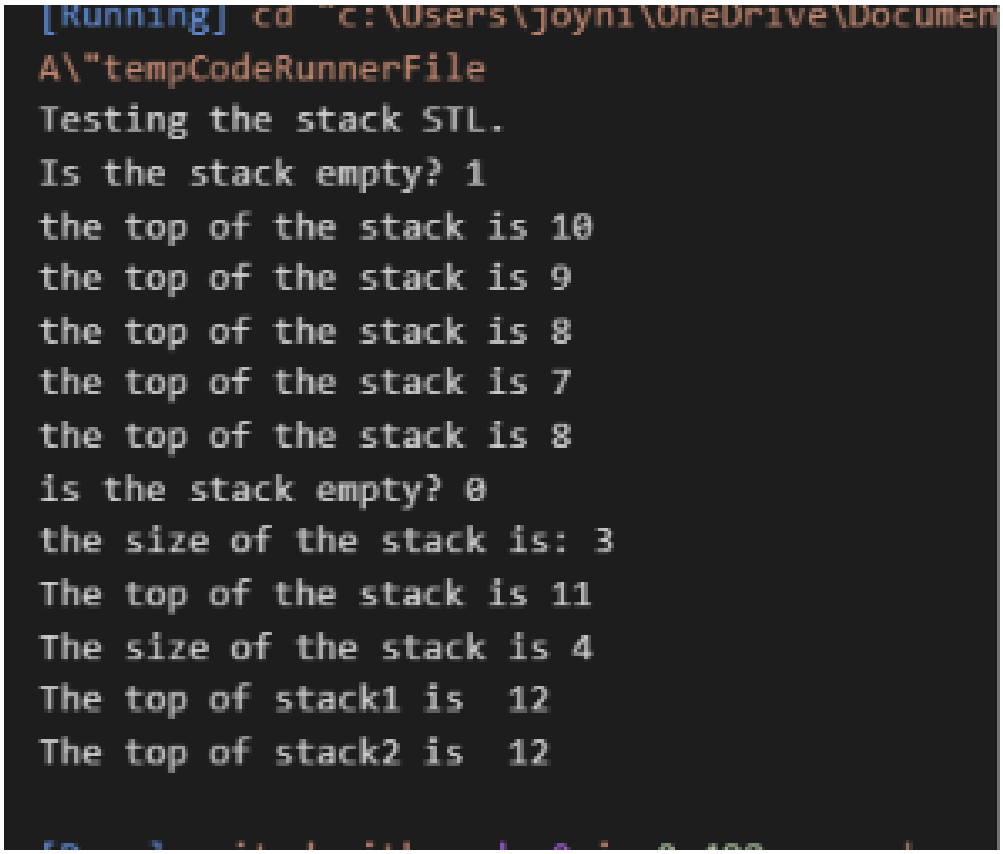
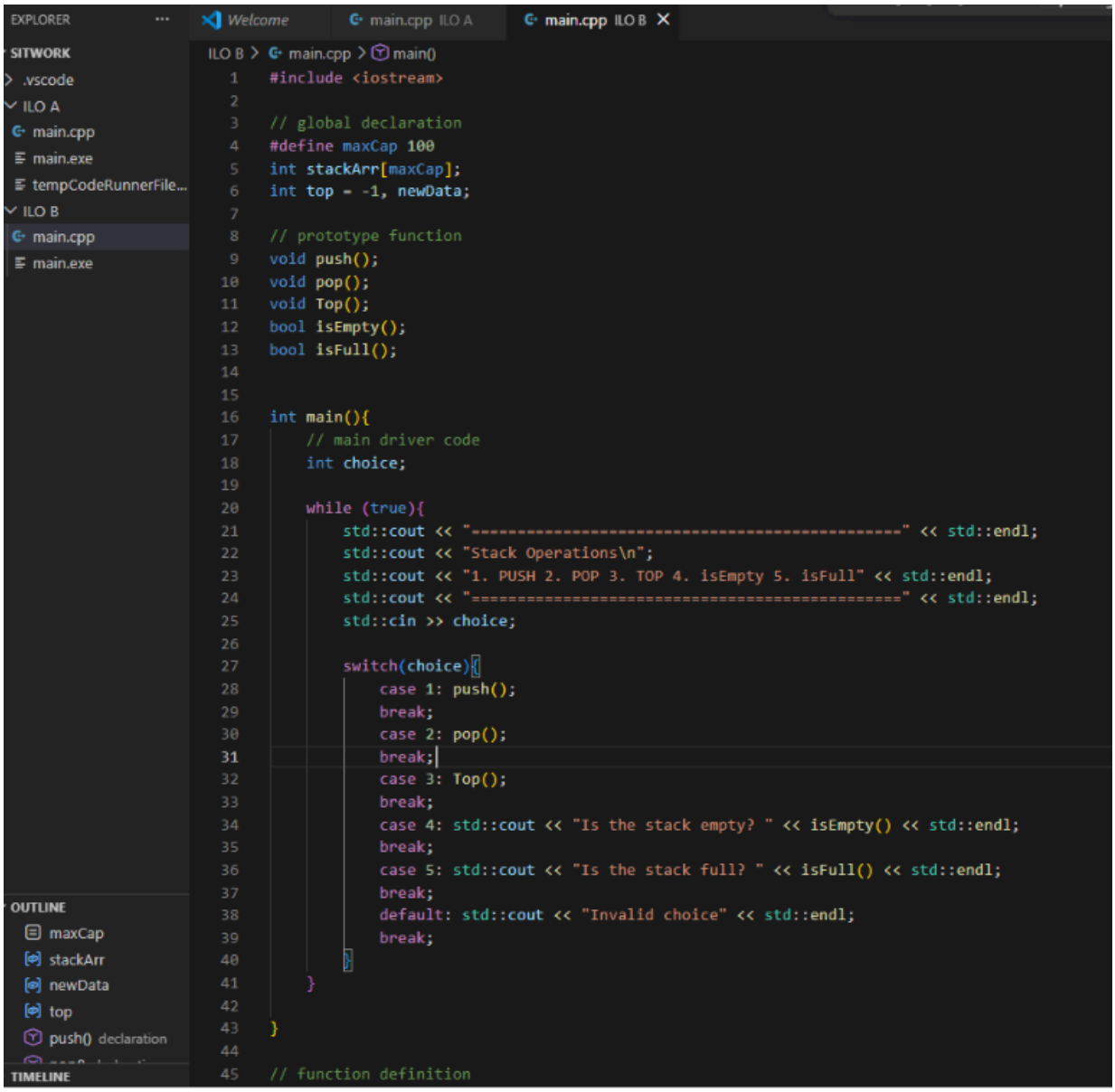


Figure 1. Code Structure of ILO A: C++ STL Stacks.

This code contains the basic ways of manipulating stacks in the C++ language using the stack library, which contains a simple template of the implementation of the core function of pushing and popping stacks, and also other functions like the emplace, empty, and swap functions. These functions from the STL library serve as the standard way of implementing stack management, but it is not customizable by the coder other than the functions provided by the library. In terms of how this code works, it first declares the variables for stack1 and stack2, which is used later for swapping. The first values (10, 9, 8, 7) are pushed to the STL stack using the push function with the parameter containing the four respective values. The pop function automatically takes out the latest element of the stack and reduces the number of the stack as well. When the number of the stacks is displayed, the number is subtracted from the number of the total stacks before. Moving over to the swap, this process involves pushing one value to another stack, so it can be used as the value for swapping. Then, it executes a swap function swapping the value of stack2 from stack1.


EXPLORER
Welcome
main.cpp ILO A
main.cpp ILO B
SITWORK
.vscode
ILO A
main.cpp
main.exe
tempCodeRunnerFile...
ILO B
main.cpp
main.exe
OUTLINE
maxCap
stackArr
newData
top
push() declaration
TIMELINE
ILO B > main.cpp > main()
#include <iostream>
// global declaration
#define maxCap 100
int stackArr[maxCap];
int top = -1, newData;
// prototype function
void push();
void pop();
void Top();
bool isEmpty();
bool isFull();
int main(){
// main driver code
int choice;
while (true){
std::cout << "==========" << std::endl;
std::cout << "Stack Operations\n";
std::cout << "1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull" << std::endl;
std::cout << "==========" << std::endl;
std::cin >> choice;
switch(choice){
case 1: push();
break;
case 2: pop();
break;
case 3: Top();
break;
case 4: std::cout << "Is the stack empty? " << isEmpty() << std::endl;
break;
case 5: std::cout << "Is the stack full? " << isFull() << std::endl;
break;
default: std::cout << "Invalid choice" << std::endl;
break;
}
}
}
// function definition

```
bool isEmpty(){
    // check the top if it is -1
    if(top == -1) return true;
    return false;
}

bool isFull(){
    // how do we verify if the stack is full
    if(top == maxCap - 1) return true;
    return false;
}

void push(){
    // error checking
    if(isFull()){
        std::cout << "Stack Overflow" << std::endl;
        return;
    }

    // pushing to the stack
    std::cout << "Enter a new value" << std::endl;
    std::cin >> newData;

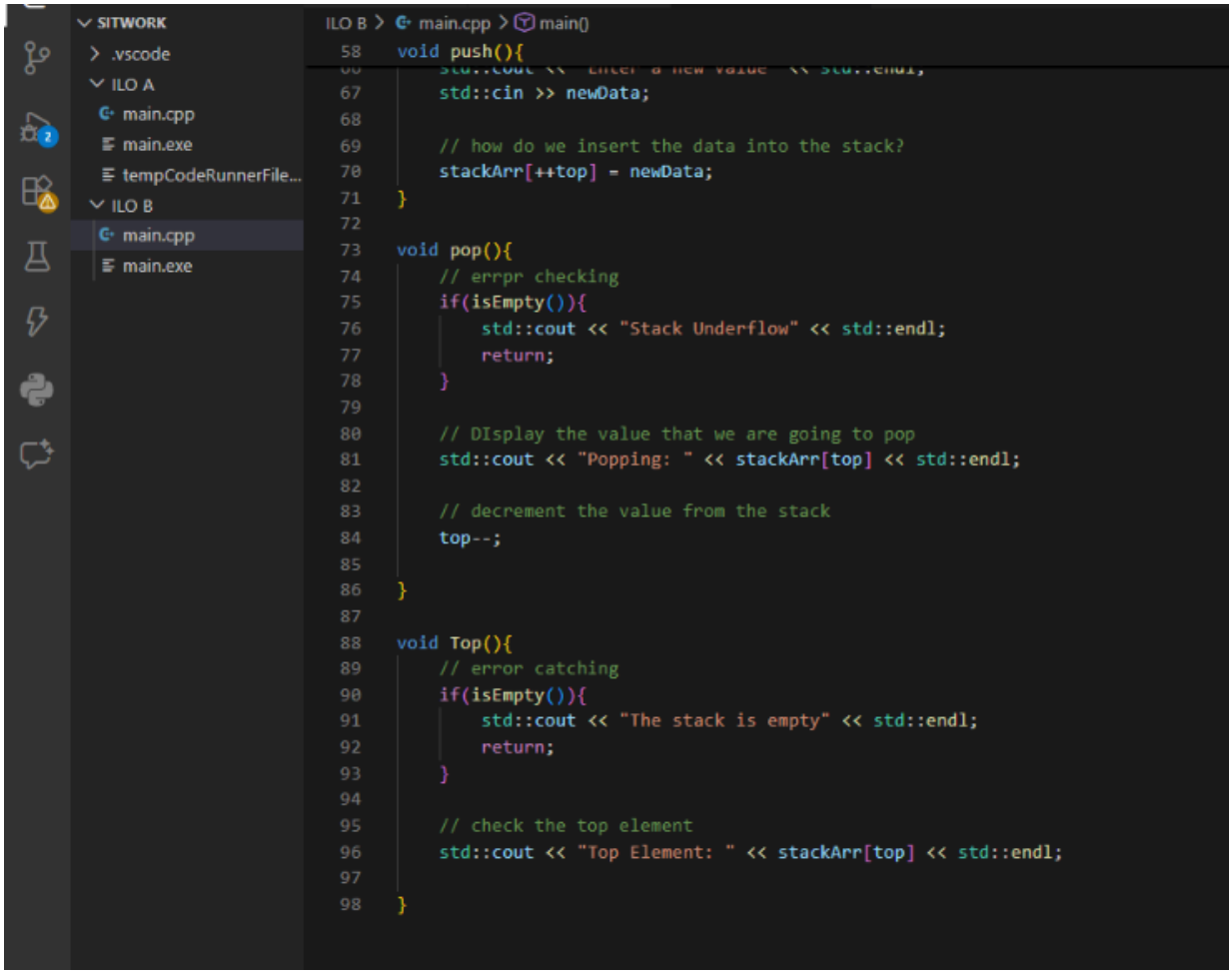
    // how do we insert the data into the stack?
    stackArr[++top] = newData;
}

void pop(){
    // errpr checking
    if(isEmpty()){
        std::cout << "Stack Underflow" << std::endl;
        return;
    }

    // DIsplay the value that we are going to pop
    std::cout << "Popping: " << stackArr[top] << std::endl;

    // decrement the value from the stack
    top--;

}
```


SITWORK
.vscode
ILO A
main.cpp
main.exe
tempCodeRunnerFile...
ILO B
main.cpp
main.exe
ILO B > main.cpp > main()
58 void push(){
67 std::cin >> newData;
68
69 // how do we insert the data into the stack?
70 stackArr[++top] = newData;
71 }
72
73 void pop(){
74 // errpr checking
75 if(isEmpty()){
76 std::cout << "Stack Underflow" << std::endl;
77 return;
78 }
79
80 // DIsplay the value that we are going to pop
81 std::cout << "Popping: " << stackArr[top] << std::endl;
82
83 // decrement the value from the stack
84 top--;
85
86 }
87
88 void Top(){
89 // error catching
90 if(isEmpty()){
91 std::cout << "The stack is empty" << std::endl;
92 return;
93 }
94
95 // check the top element
96 std::cout << "Top Element: " << stackArr[top] << std::endl;
97
98 }

OUTPUT

```
n-uhqm4cr5.kx2' '--stdout=Microsoft-MIEngine-Out-s1lmsome.4wt' '--stderr=Microsoft-MIEngine-Error-t5lceeh
' '--pid=Microsoft-MIEngine-Pid-xh424pcm.tv2' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
==========================================
Stack Operations
1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull
==========================================
4
Is the stack empty? 1
==========================================
Stack Operations
1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull
==========================================
1
Enter a new value
2
==========================================
Stack Operations
1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull
==========================================
3
Top Element: 2
==========================================
Stack Operations
1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull
==========================================
2
Popping: 2
==========================================
Stack Operations
1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull
==========================================
3
The stack is empty
==========================================
Stack Operations
1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull
==========================================
4
Is the stack empty? 1
==========================================
Stack Operations
1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull
==========================================
3
The stack is empty
==========================================
Stack Operations
1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull
==========================================
```

This program demonstrates a basic implementation of the Stack data structure using a fixed-size array in C++. A stack follows .program, which consists of a global declaration, prototype function, function definition, and driver code. The main driver code contains the user input menu for using this program's features, which are isEmpty, isFull, top, push, and pull. These functions perform the basic operations done inside a stack, including error catching to detect a stack overflow and underflow.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google , gemini to guide me and to explain give clarification**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **To guide me also from language**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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